

Checkpoint Assessment – Excel Analytics & Statistical Analysis

Problem Statement: EV Charging Network Usage Analysis

Build an **Excel-based analytical report** using the **EV Charging Network Usage dataset**. The goal is to evaluate your ability to perform **data cleaning, transformation, statistical analysis, and dashboard development using Excel**.

Objective

Create an Excel-based analysis to:

- Clean and prepare the EV charging dataset
 - Perform **descriptive statistical analysis**
 - Apply Excel formulas (logical, lookup, text, math, statistical)
 - Perform **industry-focused analysis of EV charging usage**
 - Use Pivot Tables to summarize charging behavior
 - Create charts to visualize insights
 - Build an **interactive dashboard summarizing operational insights**
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Time Allotment

2 Hours

Prerequisites

Students should be familiar with:

- Importing CSV datasets into Excel
- Excel formulas:
 - Logical functions
 - Lookup functions
 - Statistical functions
 - Arithmetic functions
- Data cleaning techniques
- Pivot Tables and Pivot Charts
- Conditional Formatting
- Excel dashboard creation (Slicers, KPIs, Charts)

Dataset Description

The dataset represents **electric vehicle charging sessions across multiple cities**.

Column	Description
Session_ID	Unique charging session identifier
City	City where charging occurred
Station_Type	Type of charging station
Vehicle_Type	EV type (2W, 3W, 4W, Bus)

Membership_Type	Member or Non-member
Charging_Start	Charging start time
Charging_End	Charging end time
Charging_Duration_Minutes	Total charging duration
Energy_Consumed_kWh	Electricity consumed
Price_per_kWh	Price charged per kWh
Revenue	Revenue generated
Payment_Mode	Mode of payment
Customer_Rating	Customer satisfaction rating

Dataset Link : [EV Charging Network Usage](#)

Functional Requirements

1. Data Loading

- Import the dataset into Excel.
 - Convert the dataset into an **Excel Table**.
 - Ensure proper column formatting:
 - Dates
 - Numeric fields
 - Text fields
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2. Data Cleaning

Perform the following cleaning tasks:

- Check for **missing values**
- Identify and remove **duplicate records**
- Ensure numeric columns contain **valid numerical values**
- Verify date and time formats
- Validate the revenue calculation

Example validation:

Revenue = Energy_Consumed_kWh × Price_per_kWh

3. Data Transformation

Create the following calculated columns using Excel formulas.

Charging Time Category

Using IF function classify charging duration:

Duration	Category
< 30 min	Short Charging
30–60 min	Medium Charging
> 60 min	Long Charging

Energy Usage Level

Create a new column **Energy_Level**

Energy Consumed	Category
< 10 kWh	Low
10–30 kWh	Medium
> 30 kWh	High

Customer Satisfaction Level

Using **Customer_Rating** classify:

Rating	Category
1–2	Low Satisfaction
3–4	Medium Satisfaction
5	High Satisfaction

4. LOOKUP Tasks

1. Create a lookup summary table using UNIQUE, AVERAGEIF.

Example table structure:

City	Average Energy Consumption	Average Charging duration
City Delhi		
City Bangalore		
City Ahmedabad and so on		

2. Perform Lookup Analysis

Create a separate lookup table and retrieve the **average energy consumption** and **average charging duration** for the following cities:

City	Avg Energy Consumption	Avg Charging Duration
Bangalore		
Mumbai		
Delhi		
Chennai		
Hyderabad		

Use **VLOOKUP** to fetch the values from the summary table created earlier.

5. Statistical Analysis

Task: Perform Descriptive Statistical Analysis

Task 1 : Using Excel statistical functions, analyze the **following** :

1. What is the average energy consumed per charging session?
2. Which station type is used most frequently? (Hint : Use **INDEX-MATCH** and **MODE** for text values)
3. Is there high variation in revenue generated from charging sessions?
4. Compare the variance of **Energy_Consumed_kWh** and **Price_per_kWh**. Which variable shows greater variability and why?

Task 2 : Calculate correlation between:

- Charging Duration and Energy Consumed
- Energy Consumed and Revenue

6. Data Analysis (Pivot Tables)

Create pivot tables to answer the following:

a) Station Utilization

- Count of charging sessions by Station_Type
 - Average Charging Duration by Station_Typ
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b) Vehicle Usage

- Energy consumption by Vehicle_Type
 - Revenue generated by Vehicle_Type
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c) Customer Experience

- Average Rating by Station_Type
 - Rating distribution by Membership_Type
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7. Visualization Requirements

Create the following charts.

Charts Required

1. Bar Chart

Revenue by City

2. Column Chart

Average Energy Consumption by Vehicle Type

3. Scatter Plot

Charging Duration vs Energy Consumption

4. Pie Chart

Distribution of Station Types

8. Conditional Formatting

Apply the following rules:

- Highlight **Revenue in Top 10%**
- Highlight **Charging Duration > 90 minutes**
- Apply **color scale to Customer_Rating**

9. Dashboard Creation

Create an **Excel Dashboard** including:

KPI Metrics

- Total Revenue

- Total Charging Sessions
- Average Energy Consumed
- Average Customer Rating

Dashboard Charts

Include:

- Revenue by City
- Vehicle Type Usage
- Energy Consumption Trends (**Hint : Extract Date and Time in Separate Columns using DATE Function**)
- Station Type Distribution

Interactivity

Add **Slicer** for:

- City
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Technical Constraints

- Only **Microsoft Excel** allowed
 - No Python / Power BI / Macros
 - Analysis must be **formula-driven or pivot-based**
 - Maintain clean sheet structure
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Submission Guidelines

Students must submit:

An Excel file containing:

- Raw dataset sheet
- Cleaned dataset sheet
- Calculated columns
- Lookup tables
- Statistical analysis sheet
- Pivot tables
- Visualizations
- Final dashboard

Ensure the workbook:

- Contains **no broken formulas**
 - Uses **clear sheet names**
 - Is **properly formatted and readable**
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